

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

16
20
20/11

Annexure A

Industry Problem Solving Task

Code of Conduct:

I T. Kasimulla / 192065046 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kasimulla

① Network Design

In a star topology, all devices are connected to a central switch. For a 20-person startup, a 24-port managed switch can be used so that there are additional ports for future expansion.

Network Components

Device	Purpose
24-port switch	Connects all devices
Router/Firewall	Internet and security
Cat6 UTP cable	Connects computers
Wifi Access point	Wireless connectivity
Computers/Printers	End devices

Advantages:

- Easy to manage
- Easy to troubleshoot
- Failure of one cable affects only one device
- Easy to add new computers
- Provides good performance

Conclusion

Star topology is a suitable choice for a 20-person startup because it is simple, reliable, manageable and scalable.

② Partial Mesh topology

A bank requires high availability because network failure can affect important financial services. A partial mesh can provide multiple paths between important network devices.

Network Components:

Component	Purpose
Core switches	High-Speed network communication
Routers	Internet/WAN Connectivity
Firewalls	Network security
Fiber optic cable	High-Speed core connections
UPS	Backup power

Features

- Multiple communication paths
- High reliability
- Reduced downtime
- Automatic alternative paths
- Suitable for critical systems
- OSPF or BGP can be used for routing.

Conclusion:

A partial mesh provides the redundancy and reliability required by a banking network.

Bus Topology

A bus topology is suitable for a small classroom when the main requirement is low cost. All computers share a common communication line. Traditional implementations can use RG-58 coaxial cable, T-connectors, BNC connectors and terminators.

Components

Component	Purpose
Coaxial cable	Main backbone
T-Connector	Connects computers
BNC Connector	Cable connection
Terminator	Prevents signal reflection
Computers	End devices

Advantages

- Low hardware cost
- Simple design
- Suitable for small networks
- Requires less equipment

Disadvantages:

- Backbone failure can affect the whole network
- Collisions can occur
- Difficult to expand
- Lower reliability compared with modern switched networks

Conclusion

A VLAN

Conclusion

Bus topology is suitable when cost is at consideration and network traffic is low

④ Hybrid Topology

A large university has many buildings and thousands of users, so a combination of topology is more suitable than using only one topology.

Network Structure:

Area

Topology/Technology

Campus Backbone Ring/Partial Mesh

Building Network Star

Laboratory Bus/Tree where suitable

Building Connection Fiber Optic

User Devices Cat6

Main Features:

- Fiber backbone
- Redundant connections
- Star topology inside buildings
- Core, distribution and access layers
- PoE support for wireless access points
- Easy Expansion